AI-Driven Protein Design with Diffusion and Language Models

Abstract

The field of protein design has undergone a paradigm shift from physics-based energy minimization to data-driven generative modeling, driven by the convergence of deep generative architectures and massive biological datasets. This survey comprehensively reviews the state-of-the-art in AI-driven protein design, focusing on two dominant methodological families: diffusion probabilistic models and protein language models (pLMs). We analyze the evolution from sequence-only autoregressive generation to joint sequence-structure co-design frameworks that leverage SE(3)-equivariant diffusion, discrete masked diffusion, and flow matching. We categorize existing approaches into backbone generation, inverse folding, sequence-structure co-generation, and functional conditioning, highlighting innovations such as latent space compression, retrieval-augmented generation, and reinforcement learning alignment. Furthermore, we examine the integration of these models into agentic workflows for multi-objective optimization and the emerging challenges in evaluating designability, diversity, and experimental validity. By synthesizing evidence from over 300 recent studies, this paper provides a technical taxonomy of the field, identifying key trends toward all-atom generation, dynamic ensemble modeling, and the unification of geometric and semantic protein representations.

1. Introduction

Protein design—the inverse problem of determining amino acid sequences that fold into specific three-dimensional structures with desired functions—has historically relied on computational methods rooted in thermodynamics and statistical mechanics, such as Rosetta [1]. While effective, these physics-based approaches are often computationally prohibitive and limited by the accuracy of energy functions in capturing complex solvation and entropic effects. The advent of deep learning has revolutionized this landscape, transitioning the field toward generative models that learn the underlying probability distributions of natural proteins from vast repositories like the Protein Data Bank (PDB) and UniProt [2, 3].

The current era of protein design is defined by the synergy between two powerful classes of models: Protein Language Models (pLMs) and Diffusion Models. pLMs, adapted from natural language processing, treat amino acid sequences as tokens, learning evolutionary constraints and "grammatical" rules of protein folding from billions of sequences [4, 5]. Concurrently, diffusion models, inspired by non-equilibrium thermodynamics, have emerged as the gold standard for generating 3D structures, offering superior sample quality and mode coverage compared to Generative Adversarial Networks (GANs) or Variational Autoencoders (VAEs) [6, 7]. Recent advancements have moved beyond treating sequence and structure as separate modalities, fostering the development of co-generative frameworks that simultaneously optimize both to ensure physical plausibility and functional efficacy [8, 9].

This survey provides a comprehensive analysis of AI-driven protein design, structured around the methodological innovations enabled by diffusion and language models. We begin by establishing a taxonomy of generative architectures, distinguishing between discrete sequence modeling, continuous geometric diffusion, and hybrid co-design systems. We then delve into specific technical mechanisms, including equivariant neural networks for 3D consistency, latent space compression for efficiency, and conditioning strategies for functional control. Finally, we discuss the integration of these models into agentic systems for iterative optimization and address critical challenges regarding evaluation benchmarks, experimental validation, and the modeling of protein dynamics.

2. Method

The methodological landscape of AI-driven protein design can be organized into four primary dimensions: (1) Sequence-centric generation using language models, (2) Structure-centric generation using geometric diffusion, (3) Joint sequence-structure co-design, and (4) Conditioning and optimization strategies. This section synthesizes the technical innovations across these domains.

2.1. Protein Language Models for Sequence Generation

Protein Language Models (pLMs) leverage the analogy between amino acid sequences and natural language to learn the statistical regularities of protein evolution. These models generally fall into autoregressive (AR) and masked language modeling (MLM) paradigms, with recent extensions incorporating discrete diffusion.

2.1.1. Autoregressive and Masked Modeling Foundations Early foundational models like ProGen and ProtGPT2 utilized autoregressive transformer architectures to generate sequences token-by-token, conditioned on functional tags or taxonomic identifiers [10, 4]. ProGen demonstrated that conditioning on evolutionary tags enables the generation of viable proteins with near-native conformational energies, even with lower primary sequence identity to natural homologs [4]. Similarly, RITA established scaling laws for autoregressive protein models, showing that performance gains follow a power law with model size, albeit with a steeper exponent than in NLP [11].

Parallel to AR models, masked language models like ESM-2 have become ubiquitous for representation learning and inverse folding tasks. These models predict masked residues based on bidirectional context, capturing long-range dependencies essential for structural stability [12, 5]. Innovations such as the Fill-in-Middle (FIM) transformation have adapted causal models to support bidirectional context during infilling tasks, significantly outperforming standard CLM approaches in structure conservation [13]. Furthermore, models like MSA2Prot have integrated Multiple Sequence Alignment (MSA) information directly into the generation process via axial attention, allowing for few-shot generalization to unseen protein families [14].

2.1.2. Discrete Diffusion for Sequence Design A significant recent trend is the application of discrete diffusion processes to sequence generation, bridging the gap between MLM and generative modeling. Models like DPLM (Diffusion Protein Language Model) implement discrete diffusion with absorbing state masking, generalizing both masked and autoregressive objectives to enable high-quality bidirectional generation [15]. DPLM-2 extends this to multimodal settings, jointly modeling sequences and discretized structure tokens via Lookup-Free Quantization (LFQ), demonstrating that discrete tokenization enables language models to learn joint distributions effectively [16].

Innovations in discrete diffusion include the development of specialized noise schedules and parameterizations. For instance, MeMDLM applies masked discrete diffusion specifically to membrane proteins, utilizing a SUBS-parameterized reverse process to guarantee strict token unmasking while preserving bidirectional context [17]. Similarly, SCISOR employs an insertion-only forward noising process to model deletions in the reverse process, achieving state-of-the-art performance in protein shrinking tasks [18]. Theoretical advancements have also unified discrete, Gaussian, and simplicial diffusion under the Wright-Fisher population genetics process, providing a rigorous mathematical foundation for discrete sequence modeling [19].

2.1.3. Latent Space and Continuous Representations To improve sampling efficiency and capture continuous semantic properties, several methods operate in the latent space of pretrained pLMs. DiMA (Diffusion on Model Encodings) performs diffusion in the continuous embedding space of frozen encoders like ESM-2, utilizing a Tan-10 noise schedule to adapt dynamics to protein sequentiality [12]. This approach captures structural semantics superior to discrete token modeling, though larger encoders may reduce sequence diversity [12]. ProtFlow further compresses pLM latent spaces via a transformer-based compressor and applies flow matching with reflow techniques to enable single-step generation, significantly accelerating inference while maintaining foldability [20].

2.2. Geometric Diffusion for Structure Generation

While pLMs excel at sequence modeling, generating novel 3D backbones requires architectures that respect the geometric symmetries of physical space. Diffusion models have become the dominant framework for this task, evolving from coordinate-based approaches to sophisticated frame-based and latent representations.

2.2.1. SE(3)-Equivariant Diffusion on Manifolds The core challenge in structure generation is ensuring rotational and translational equivariance. Early models like ProtDiff employed E(3)-equivariant graph neural networks to diffuse C-alpha coordinates directly [21]. However, subsequent work recognized that diffusing on the manifold of rigid body frames (SE(3)) yields superior results. FrameDiff decomposes the SE(3) diffusion process into independent SO(3) rotation and $\mathbb{R}^3$ translation components, utilizing Invariant Point Attention (IPA) to enforce equivariance [22]. This decomposition allows for tractable training on compact Lie groups and has become a standard backbone for many generative models.

Genie and Genie 2 further refined this approach by employing an asymmetric representation: injecting isotropic Gaussian noise into Cartesian coordinates during the forward process but predicting noise in the frame representation during the reverse process [23, 24]. This decoupling simplifies the diffusion objective while preserving geometric fidelity. Genie 2, trained on the massive AlphaFold Database, demonstrates the ability to scaffold multiple motifs with unspecified inter-motif positions, significantly increasing design diversity [24].

2.2.2. Flow Matching and Alternative Dynamics Flow Matching (FM) has emerged as a competitive alternative to standard Denoising Diffusion Probabilistic Models (DDPM), offering deterministic sampling trajectories and improved convergence. FOLDFLOW utilizes simulation-free conditional flow matching on SE(3) manifolds, employing Riemannian Optimal Transport to straighten probability paths and reduce training variance [25]. Proteína scales this approach using non-equivariant transformers with hierarchical fold class conditioning, achieving state-of-the-art designability rates on long chains by leveraging synthetic data scaling [26].

Innovations in dynamics include the integration of physics-inspired processes. ProT-GFDM replaces standard Brownian motion with fractional Brownian motion (fBm) to capture long-range dependencies in protein backbones, utilizing a Markov approximation for tractable inference [27]. Similarly, PhysFlow integrates non-linear Hamiltonian unfolding dynamics with SE(3) flow matching, enforcing physical realism through Coulomb repulsion and secondary structure potentials [28]. These physics-informed approaches aim to generate structures that are not only statistically plausible but also thermodynamically stable.

2.2.3. Latent and Tokenized Structure Representations To mitigate the computational cost of all-atom diffusion, several methods compress 3D structures into latent spaces or discrete tokens. LatentDiff employs an SE(3)-equivariant graph autoencoder to compress backbones into a lower-dimensional latent space before applying diffusion, significantly reducing modeling complexity [29]. SaDiT maps continuous SE(3) frames to discrete SaProt tokens via geometry-invariant encoding, operating a Diffusion Transformer on these embeddings to prevent coordinate drift [30].

Vector quantization techniques have also gained traction. FoldToken utilizes Soft Conditional Vector Quantization (SoftCVQ) to project residue types and folding angles into discrete latent codes, enabling autoregressive generation of structures with high reconstruction fidelity [31]. Yeti combines Lookup-Free Quantization (LFQ) with flow matching in a single-stage end-to-end tokenizer, achieving superior codebook utilization and token diversity compared to frozen encoder paradigms [32]. These tokenization strategies facilitate the integration of structure generation into standard language model architectures.

2.3. Joint Sequence-Structure Co-Design

The most advanced generative systems now treat sequence and structure as coupled variables, optimizing them simultaneously to ensure mutual consistency. This co-design paradigm avoids the pitfalls of sequential pipelines where a generated structure may be undesignable or a designed sequence may misfold.

2.3.1. Hybrid Discrete-Continuous Diffusion Co-design models typically employ hybrid diffusion processes that handle discrete amino acid types and continuous coordinates within a unified framework. RFDiffusion and its successors utilize joint diffusion on backbone frames and sequence logits, often leveraging pretrained inverse folding models like ProteinMPNN as guidance [33, 3]. A-CODE (Atomic CO-DEsign) advances this by integrating mask diffusion for discrete atom types with continuous coordinate diffusion in a shared transformer, predicting atom names directly to infer residue identities and enabling the design of non-canonical amino acids [34].

Discrete diffusion frameworks have also been extended to co-design. DPLM-2 and HD-Prot employ hybrid diffusion objectives that mask both sequence tokens and continuous (or quantized) structure tokens simultaneously [35, 16]. HD-Prot specifically uses non-quantized continuous structure tokens to preserve fine-grained geometric details while maintaining compatibility with discrete sequence modeling [35]. CodeFP utilizes a co-generative discrete diffusion framework to jointly decode amino acid sequences and quantized local structure tokens, integrating a Functional-Structural Retrieval module to inject hierarchical inductive biases [9].

2.3.2. All-Atom and Side-Chain Modeling Moving beyond C-alpha backbones, recent models aim for full all-atom generation. SeedProteo adapts AlphaFold3-like architectures for generative tasks, utilizing an atom14 representation with virtual atoms and a global Markov Random Field (MRF) decoder to capture high-order residue dependencies [36]. AbDiffuser employs an Aligned Protein Mixer (APMixer) operating on fixed-length AHo-numbered sequences to generate full-atom antibody structures, enforcing coordinate-space projections to guarantee physical validity [37].

Side-chain packing, a critical component of all-atom design, is addressed by models like DiffPack, which employs autoregressive torsional diffusion on chi angles to mitigate cumulative perturbation errors [38]. Similarly, SidechainDiff utilizes conditional Riemannian diffusion on a 4-dimensional torus to generate side-chain rotamer distributions, capturing interface uncertainty better than deterministic methods [39]. These approaches ensure that generated backbones are populated with sterically feasible side chains, a prerequisite for experimental realization.

2.3.3. Motif Scaffolding and Inpainting A key application of co-design is motif scaffolding, where a functional motif must be embedded within a stable scaffold. Floating Anchor Diffusion (FADiff) treats functional motifs as rigid entities that float independently during the diffusion process, automatically discovering optimal spatial arrangements without fixed inter-motif coordinates [40]. ProtPainter enables precise topology control by allowing users to sketch 3D curves that guide the backbone generation, bridging coarse-grained design intent with atomic-level realization [41].

Inpainting methods fill missing regions of a structure or sequence. DiffSDS inserts an Atomic Direction Space (ADS) into a language transformer to efficiently impose geometric constraints during backbone inpainting [42]. Bridge-IF utilizes Markov bridges to learn the probabilistic dependency between structure-derived priors and native sequences, mitigating exposure bias in inverse folding [43]. These tools allow for the modular engineering of proteins by preserving functional sites while redesigning surrounding contexts.

2.4. Conditioning, Control, and Optimization

Generating proteins with specific functions requires robust conditioning mechanisms. Methods range from simple tag concatenation to complex reinforcement learning loops and agentic planning.

2.4.1. Multimodal and Text-Based Conditioning Natural language provides a flexible interface for specifying design goals. InstructPro utilizes shared latent memory vectors to distill semantic features from natural language instructions, enabling instruction-guided design without explicit 3D structures [44]. ProtDAT integrates protein sequences and descriptive texts via a Multimodal Cross-attention Mechanism, stabilizing autoregressive generation through fine-grained text-sequence fusion [45]. For more precise control, TaxDiff incorporates biological taxonomy identifiers as explicit guidance signals at every transformer layer, enabling fine-grained specification of protein properties [46].

Multimodal conditioning also extends to ligand and functional annotations. ProtLiD2 integrates explicit ligand conditioning into discrete sequence-structure diffusion via geometry-aware cross-attention, allowing for ligand-aware co-design [47]. CFP-GEN employs Annotation-Guided Feature Modulation (AGFM) to compose functional tags (GO, IPR, EC) and sequence motifs, enabling precise multi-objective design [48].

2.4.2. Reinforcement Learning and Preference Optimization To align generative models with specific objectives like stability or binding affinity, Reinforcement Learning (RL) and Direct Preference Optimization (DPO) are increasingly employed. DPO_pLM adapts DPO to autoregressive pLMs, treating sequences as actions and oracle scores as rewards to align models with external objectives without model collapse [49]. Physio-DPO integrates a magnitude-aware weighting function into the DPO loss, amplifying penalties for severe thermodynamic violations to eliminate structural hallucinations [50].

RL frameworks like ProDCARL fine-tune diffusion priors on specific domains (e.g., antimicrobial peptides) and couple generators with classifiers for activity and toxicity rewards, optimizing policies via top-k policy gradient updates [51]. Similarly, AAVGen utilizes Group Sequence Policy Optimization (GSPO) to navigate the multi-objective landscape of viral capsid design, balancing production fitness, tropism, and thermostability [52]. These methods enable the steering of generative distributions toward regions of the fitness landscape that are underrepresented in training data.

2.4.3. Agentic and Tool-Augmented Workflows The complexity of protein design has spurred the development of agentic systems that decompose tasks into planning, execution, and reflection. ProtoCycle couples an LLM planner with a lightweight tool environment, enabling iterative tool-augmented cycles where the agent reflects on feedback scores to replan strategies [53]. VibeGen employs a dual-agent architecture comprising a Protein Designer and a Protein Predictor to map between sequence space and vibrational normal mode shapes, facilitating the design of proteins with tailored dynamics [54].

Decentralized swarm intelligence approaches assign individual LLM agents to specific residue positions, allowing emergent sequence design through local neighborhood interactions and global memory [55]. RosettaSearch deploys LLMs as generative optimizers within a priority-based parallel search algorithm, using structured structural feedback to guide mutations [56]. These agentic frameworks mimic human engineering workflows, leveraging the reasoning capabilities of LLMs to navigate complex design spaces.

2.5. Retrieval-Augmented and Evolutionary Approaches

Incorporating evolutionary information and retrieving homologous structures can significantly enhance design quality, especially in data-scarce regimes.

2.5.1. Retrieval-Augmented Generation (RAG) RAG methods retrieve structurally or functionally similar proteins to condition the generative process. RadDiff executes a hierarchical search using FoldSeek and US-align to retrieve homologous proteins, constructing position-specific amino acid profiles to guide inverse folding [57]. RADAb employs a structure-informed retrieval module to query the PDB for CDR-like fragments, integrating these motifs into a dual-branch denoising network to enhance antibody design [58]. RADiAnce constructs a shared contrastive latent space for binding sites and binders, enabling accurate cross-domain retrieval without explicit binder templates [59].

2.5.2. Evolutionary Modeling and Indels Modeling insertions and deletions (indels) is crucial for capturing evolutionary diversity. DPLM-Evo decouples variable-length observed sequence space from an upsampled latent alignment space containing gap tokens, rendering indel modeling tractable within discrete diffusion frameworks [60]. EvoFlows utilizes continuous-time Markov chain edit flows to learn mutational trajectories between evolutionarily related sequences, natively supporting variable-length generation [61]. ProfileBFN extends Bayesian Flow Networks to accept MSA profiles as input, treating single sequences as degenerate profiles to enable scalable generative modeling of protein families [62].

3. Challenges and Outlook

Despite rapid progress, several critical challenges remain in AI-driven protein design.

3.1. Evaluation and Benchmarking

A major bottleneck is the lack of standardized, comprehensive benchmarks. While metrics like pLDDT, TM-score, and sequence recovery are widely used, they often correlate poorly with experimental success [8, 63]. ProteinBench and PDFBENCH attempt to address this by establishing holistic evaluation frameworks covering quality, novelty, diversity, and semantic alignment, yet discrepancies between in silico metrics and wet-lab results persist [64, 65]. The "designability" metric, often assessed via AlphaFold2 confidence scores, can be misleading if the predictor is biased toward the training distribution of the generator [66]. Furthermore, evaluating functional properties like binding affinity or enzymatic activity remains reliant on computationally expensive docking or molecular dynamics simulations, which themselves have limited accuracy [67, 68].

3.2. Dynamics and Conformational Ensembles

Most current models generate static structures, failing to capture the dynamic conformational ensembles essential for protein function, allostery, and signaling [1, 69]. While methods like EigenFold and AlphaFolding attempt to model conformational landscapes via harmonic oscillators or 4D diffusion, they struggle with large-scale transitions and millisecond-to-second dynamics [70, 71]. Generative models for intrinsically disordered proteins (IDPs) like IDPForge and machine learning potentials for ensemble generation are emerging but remain less mature than folded protein designers [72, 73]. Future research must prioritize the generation of thermodynamic ensembles rather than single energy minima.

3.3. Data Scarcity and Bias

Generative models are heavily reliant on the PDB and UniProt, which are biased toward soluble, stable, and easily crystallized proteins [3, 24]. This bias limits the exploration of novel folds, membrane proteins, and complex assemblies. Synthetic data generation, as employed by Proteína and Teddymer, offers a pathway to expand the training distribution, but risks reinforcing model biases if not carefully curated [26, 74]. Additionally, the scarcity of labeled functional data (e.g., kinetic parameters, binding constants) hinders the supervised fine-tuning of models for specific functions [75, 76].

3.4. Safety and Biosecurity

The ability to generate novel proteins raises significant biosecurity concerns. While models like ProGen and DPLM can theoretically generate toxic or pathogenic sequences, current safeguards are limited [2, 66]. The dual-use nature of these technologies necessitates the development of robust screening mechanisms and ethical guidelines to prevent the malicious design of biological agents. Research into "safe-by-design" constraints and detectable watermarks in generated sequences is urgently needed [77].

3.5. Future Directions

Future research will likely focus on the unification of sequence, structure, and function into single foundation models capable of "any-to-any" generation [78]. The integration of physics-based simulations with generative priors, such as force-guided sampling and differentiable molecular dynamics, will enhance the physical realism of designs [79, 80]. Moreover, the development of autonomous agentic systems that can plan, execute, and validate design cycles in closed-loop experimental setups represents the next frontier in automated protein engineering [53, 81]. Finally, extending these methods to non-protein biomolecules, including nucleic acids and synthetic polymers, will broaden the impact of generative AI in molecular design [82, 83].

4. Conclusion

AI-driven protein design has evolved from simple sequence modeling to sophisticated, geometry-aware generative frameworks capable of designing novel proteins with atomic precision. The convergence of diffusion models and protein language models has enabled unprecedented control over protein structure and function, facilitating applications in therapeutics, enzymology, and materials science. While challenges in evaluation, dynamics modeling, and data bias remain, the rapid pace of innovation suggests that fully autonomous, physics-consistent protein design pipelines are within reach. As these technologies mature, they promise to unlock new frontiers in biological understanding and engineering, transforming the way we design and interact with the molecular world.

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